Towards Reliable Machine Learning: Theory and Practice

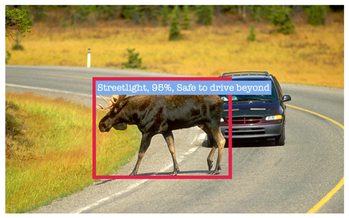
Sean Du, Assistant Professor, NTU, xuefeng.du@ntu.edu.sg

w/ Prof. Zhen Fang[1], Ilias Diakonikolas[2], Chaowei Xiao[2], Jerry Zhu[2], Robert Nowak[2], Sharon Li[2]

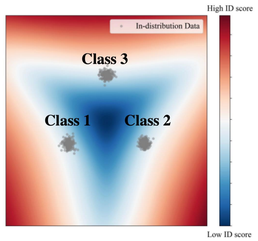
[1]University of Technology Sydney, [2]UW-Madison

More about my research

AI cannot Handle Unknowns



Overconfident predictions



Ill-fated uncertainty surface

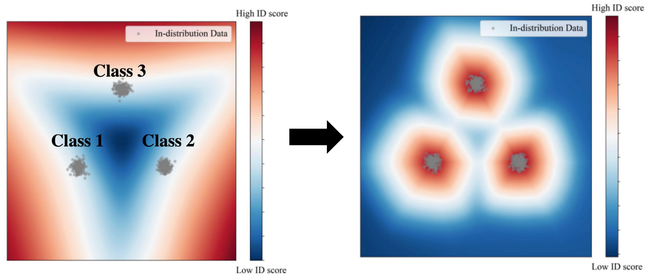
$$\mathcal{R}_{\text{in}}(f) = \frac{1}{n}\sum_{i=1}^{n} \ell\left(f\left(\mathbf{x}_i\right), y_i\right)$$

$$f^* = \text{argmin}_{f\in\mathcal{F}}\, \mathcal{R}_{\text{in}}\left(f\right)$$

Empirical Risk Minimization (ERM)

How reliable the current AI systems are when being deployed in the real world?

My Research Overview

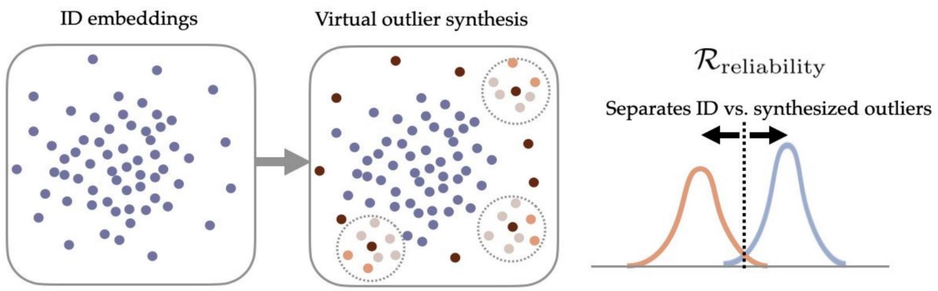


$\text{argmin}_{f\in\mathcal{F}}\, \mathcal{R}_{\text{in}}\left(f\right)$ $\quad$ $\text{argmin}_{f\in\mathcal{F}}\left[\mathcal{R}_{\text{in}}\left(f\right) + \alpha\cdot \boxed{\mathcal{R}_{\text{reliability}}\left(f\right)}\right]$

My research focuses on teach machine learning models **what they don't know**

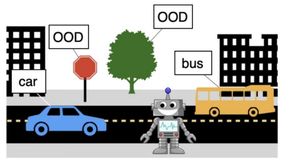
Building Unknow—Aware Learning Frameworks for Out-of-distribution Detection

Virtual Outlier Synthesis



Synthesize virtual outliers—data samples that resemble real unknown data—without needing any formation of unknowns

Leveraging Unlabeled Data

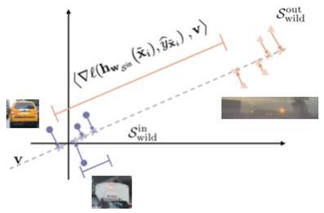


Data configuration:

- Labeled in-distribution data: $\mathcal{S}^{\text{in}} = \{(\mathbf{x}_1, y_1), \ldots, (\mathbf{x}_n, y_n)\}$ sampled from $\mathbb{P}_{\mathcal{X}\mathcal{Y}}$
- Unlabeled wild data: $\mathcal{S}_{\text{wild}} = \{\tilde{\mathbf{x}}_1, \ldots, \tilde{\mathbf{x}}_m\}$ sampled from $\mathbb{P}_{\text{wild}}$

And, inspired by WOODS [1]

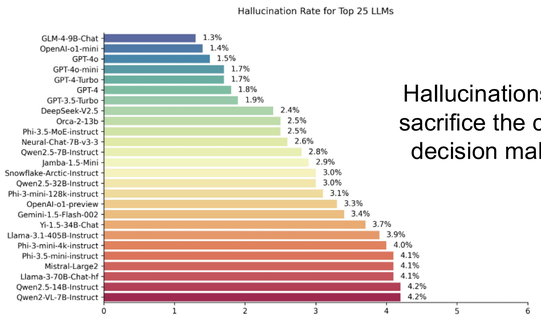
$$\mathbb{P}_{\text{wild}} := (1-\pi)\mathbb{P}_{\text{in}} + \pi\mathbb{P}_{\text{out}}$$



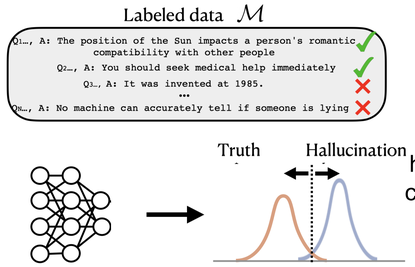
Theorem 1. (Du et al., 2024, informal). Under proper conditions, with a high probability, there exist some specific directions **(e.g., the top singular vector direction)** where **the mean magnitude of the gradients for the wild outlier data is larger than that of ID data.**

Harnessing in the wild data—data samples that can contain a mixture of in-distribution and OOD—which are freely collectable and closer to test-time OOD distribution

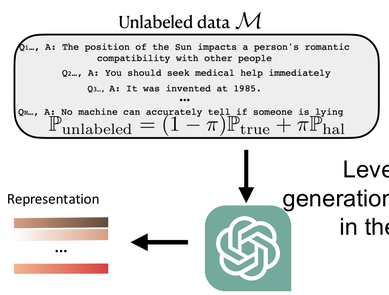
Towards Responsible Foundation Models



Hallucinations can sacrifice the critical decision making!

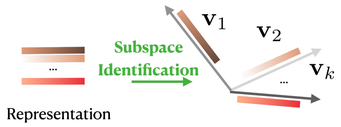


Training a supervised hallucination classifier is costly and not adaptable!



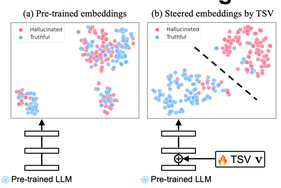
Leveraging unlabeled LLM generations for hallucination detection in the representation space

PCA projection



Du et al., NeurIPS 2024 Spotlight

Latent Steering



Park, Du et al., arXiv 2025